

A Four-Parameter Polynomial Parametrization for $xyz + ts + 1 = 0$

Abstract

We give an explicit four-parameter polynomial family of integer solutions to $xyz + ts + 1 = 0$. The construction comes from a two-factor identity for $1 + xyz$. Over any field, the resulting map is an isomorphism on the open set where $xy \neq 0$; thus it is a birational parametrization with the expected number of parameters. We also characterize exactly which integral points are obtained from integral parameters.

1 The polynomial family

Theorem 1. Fix $\varepsilon \in \{1, -1\}$. For arbitrary parameters $a, b, m, n \in \mathbb{Z}$, define

$$\begin{aligned} x &= a, & y &= b, \\ z &= \varepsilon(m + n) + abmn, & t &= \varepsilon + abm, \\ & & s &= -\varepsilon - abn. \end{aligned} \tag{1}$$

Then

$$xyz + ts + 1 = 0.$$

Equivalently, the five polynomials in (1) satisfy the identity $xyz + ts + 1 \equiv 0$ in $\mathbb{Z}[a, b, m, n]$.

Proof. Put $q = ab$. Since $\varepsilon^2 = 1$,

$$(\varepsilon + qm)(\varepsilon + qn) = 1 + \varepsilon q(m + n) + q^2 mn = 1 + q(\varepsilon(m + n) + qmn). \tag{2}$$

Under (1), the last expression is $1 + xyz$, whereas

$$ts = -(\varepsilon + qm)(\varepsilon + qn).$$

Thus $ts = -(1 + xyz)$, which is equivalent to the required equation. □

Taking $\varepsilon = 1$ gives the particularly simple parametrization

$$(x, y, z, t, s) = (a, b, m + n + abmn, 1 + abm, -1 - abn), \quad a, b, m, n \in \mathbb{Z}. \tag{3}$$

For instance, setting $a = b = 1$, $m = 0$, and $n = r$ yields the infinite subfamily

$$(x, y, z, t, s) = (1, 1, r, 1, -1 - r), \quad r \in \mathbb{Z}.$$

2 How the construction was found

The equation may be rewritten as

$$1 + xyz = -ts. \tag{4}$$

Write $q = xy$. A direct way to enforce (4) is to make $1 + qz$ split into two polynomial factors. Introduce two free parameters m, n and seek a factorization of the form

$$1 + qz = (\varepsilon + qm)(\varepsilon + qn), \quad \varepsilon^2 = 1.$$

Expanding the right-hand side gives

$$1 + \varepsilon q(m + n) + q^2 mn,$$

so the factorization holds precisely when

$$z = \varepsilon(m + n) + qmn.$$

One may then take one factor to be t and the negative of the other to be s :

$$t = \varepsilon + qm, \quad s = -\varepsilon - qn.$$

Finally, writing $q = ab$ leaves $x = a$ and $y = b$ as independent parameters. This produces exactly (1).

For $\varepsilon = 1$, the central identity can also be expressed through the binary operation

$$m \oplus_q n := m + n + qmn.$$

It is designed so that

$$1 + q(m \oplus_q n) = (1 + qm)(1 + qn).$$

Thus the nonlinear-looking term $m + n + qmn$ is not guessed: it is forced by multiplying two expressions congruent to 1 modulo q .

3 Generic completeness

Let

$$V : xyz + ts + 1 = 0$$

be the affine hypersurface in $\mathbb{A}^5$, and let $\Phi_\varepsilon : \mathbb{A}^4 \rightarrow V$ be the polynomial map defined by (1). For every field K , the polynomial $xyz + ts + 1$ is irreducible in $K[x, y, z, t, s]$: viewed as a polynomial in z , it is primitive over $K[x, y, t, s]$ and linear over that ring's fraction field. Hence the nonempty open subset $xy \neq 0$ is dense in V .

Proposition 2. *Over any field K , the restriction of Φ_ε to $ab \neq 0$ is an isomorphism onto the open subset of V on which $xy \neq 0$. Its inverse is*

$$a = x, \quad b = y, \quad m = \frac{t - \varepsilon}{xy}, \quad n = -\frac{s + \varepsilon}{xy}. \quad (5)$$

In particular, Φ_ε is a birational parametrization of V .

Proof. Let $(x, y, z, t, s) \in V(K)$ with $q := xy \neq 0$, and define a, b, m, n by (5). The formulas for x, y, t, s in (1) are then immediate. Moreover,

$$\begin{aligned} q(\varepsilon(m + n) + qmn) &= \varepsilon(t - \varepsilon) - \varepsilon(s + \varepsilon) - (t - \varepsilon)(s + \varepsilon) \\ &= -ts - 1. \end{aligned}$$

Since the point lies on V , one has $-ts - 1 = xyz = qz$. Division by q therefore gives

$$z = \varepsilon(m + n) + qmn.$$

Hence (5) is indeed the inverse map. Both maps are regular on the indicated principal open subsets. $\square$

The proposition explains why four independent parameters arise naturally: one equation in five affine variables has four-dimensional generic freedom, and the construction realizes that freedom without redundancy on $xy \neq 0$.

4 Integral image and the boundary $xy = 0$

The preceding proposition is a completeness statement over a field. For integer parameters, divisibility enters because the inverse formulas in (5) need not be integral.

Proposition 3. *Let $(x, y, z, t, s) \in \mathbb{Z}^5$ satisfy $xyz + ts + 1 = 0$, and suppose $xy \neq 0$. For fixed $\varepsilon \in \{1, -1\}$, this point belongs to $\Phi_\varepsilon(\mathbb{Z}^4)$ if and only if*

$$t \equiv \varepsilon \pmod{xy}. \quad (6)$$

Equivalently, one may require $s \equiv -\varepsilon \pmod{xy}$.

Proof. If the point is obtained from integral parameters, then $t - \varepsilon = xym$ and $s + \varepsilon = -xyn$, so both congruences hold.

Conversely, assume (6). Reducing the original equation modulo xy gives $ts \equiv -1 \pmod{xy}$. Since $t \equiv \varepsilon$ and $\varepsilon^2 = 1$, it follows that $s \equiv -\varepsilon \pmod{xy}$. Hence the values of m, n in (5) are integers, and Proposition 2 shows that these parameters recover the given point. $\square$

If $xy = 0$, the equation reduces to $ts = -1$. Hence an integral solution has either $(t, s) = (1, -1)$ or $(t, s) = (-1, 1)$, while z is arbitrary. The family Φ_1 covers the first case and Φ_{-1} covers the second. Thus the two sign choices cover the entire degenerate integral locus $xy = 0$.

Remark 4. The formulas above give a polynomial family of integral solutions and a birational parametrization of the full hypersurface. They are not asserted to send integral parameters onto every integral point with $xy \neq 0$; Proposition 3 gives the exact congruence criterion for such integral lifting.